# Problem H4

## Møller Scattering: $e^-e^- \to e^-e^-$

Your First Full QED Cross Section Calculation

Everything taught from scratch — no prior QED calculation experience assumed

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# Part I
# Understanding the Problem

## §1 The Problem in Plain English

**H4 — The Problem (Reworded)**

Two electrons scatter off each other at high energy ($e^- + e^- \to e^- + e^-$). We ignore the electron mass.
**Your job:** Calculate the probability of this happening (expressed as a differential cross section $d\sigma/dt$) starting from the Feynman diagrams.
The problem has you focus on one specific piece of the calculation — the **interference term** $\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle$ — because it's the one that requires a new technique. The other pieces can be recycled from electron–muon scattering (which Griffiths already worked out).

**The Road Map**

Here is the complete pipeline for this problem. We will walk through every step:

$$\text{Feynman diagrams} \xrightarrow{1} \mathcal{M} \xrightarrow{2} |\mathcal{M}|^2 \xrightarrow{3} \langle |\mathcal{M}|^2 \rangle \xrightarrow{4} \text{Mandelstam} \xrightarrow{5} \frac{d\sigma}{dt}$$

**Step 1:** Draw diagrams → write amplitude $\mathcal{M}$.
**Step 2:** Square it → get $|\mathcal{M}|^2$ (with interference terms).
**Step 3:** Average over spins → use Casimir's trick to get traces → evaluate traces.
**Step 4:** Convert dot products → Mandelstam variables $s, t, u$.
**Step 5:** Plug into cross section formula.
Every QED problem you'll ever do follows these same five steps.

## §2 What Is a Feynman Diagram, Really?

Before we compute anything, let's make sure we understand what a Feynman diagram *does*.

**Feynman Diagrams = Instructions for Building $\mathcal{M}$**

A Feynman diagram is not just a picture. It's a **recipe**: each part of the diagram tells you to write down a specific mathematical factor. The amplitude $\mathcal{M}$ is the product of all these factors.
The ingredients ("Feynman rules") are:
**External fermion lines** (electrons going in or out):

- Incoming electron → write $u(p)$ (a 4-component column vector — the spinor)
- Outgoing electron → write $\bar{u}(p)$ (a 4-component row vector — the adjoint spinor)

**Vertices** (where a photon meets two electrons):

- Each vertex → write $ig_e\gamma^\mu$ (a $4 \times 4$ matrix times a coupling constant)

- The index $\mu$ labels which "component" of the photon you're talking about

**Internal photon line** (the virtual photon being exchanged):

- Photon propagator → write $\dfrac{-ig_{\mu\nu}}{q^2}$ (where $q$ is the photon's momentum)
- The $g_{\mu\nu}$ contracts the $\mu$ index at one vertex with the $\nu$ index at the other

Then you multiply everything together, following each fermion line **backwards** (against the arrow). The result is $\mathcal{M}$, a complex number for each choice of spins.

## §3 Why Two Diagrams? (Identical Particles)

**The Identical Particle Problem**

Suppose electron #1 (momentum $p_1$) and electron #2 (momentum $p_2$) scatter, producing electron #3 (momentum $p_3$) and electron #4 (momentum $p_4$).
**Question:** Did electron 1 become electron 3? Or did electron 1 become electron 4?
**Answer:** We can't tell! Both are possible. Quantum mechanics says we must **add the amplitudes** for both possibilities, then square the sum.
This gives two Feynman diagrams:
**Diagram 1 ("direct" / t-channel):** $p_1 \to p_3$ and $p_2 \to p_4$. The virtual photon carries momentum $q = p_1 - p_3$.
**Diagram 2 ("exchange" / u-channel):** $p_1 \to p_4$ and $p_2 \to p_3$. The virtual photon carries momentum $q = p_1 - p_4$.
Because electrons are fermions, swapping two identical fermions introduces a **minus sign** (this is Griffiths Rule 8 — it's the Pauli exclusion principle in disguise):

$$\mathcal{M}_{\text{total}} = \mathcal{M}_1 - \mathcal{M}_2$$

**Compare: Electron–Muon Scattering**

In $e\mu \to e\mu$, the outgoing particles are *different* (one electron, one muon), so there's no ambiguity about which is which. Only *one* diagram. No interference term. That's why $e\mu$ scattering is simpler.

## §4 Building the Amplitudes $\mathcal{M}_1$ and $\mathcal{M}_2$

Let's build $\mathcal{M}_1$ piece by piece using the Feynman rules.

### §4.1 Diagram 1: Step by Step

In diagram 1, electron $p_1$ scatters to $p_3$ at the top vertex, and electron $p_2$ scatters to $p_4$ at the bottom vertex.

**Top vertex (fermion line $p_1 \to p_3$):** Follow the fermion line *backwards* (from outgoing $p_3$ back to incoming $p_1$):

$$\bar{u}(p_3) \cdot (ig_e\gamma^\mu) \cdot u(p_1) = ig_e\,[\bar{u}(3)\gamma^\mu u(1)]$$

**Bottom vertex (fermion line $p_2 \to p_4$):**

$$\bar{u}(p_4) \cdot (ig_e\gamma^\nu) \cdot u(p_2) = ig_e\,[\bar{u}(4)\gamma^\nu u(2)]$$

**Photon propagator** (connecting the two vertices):

$$\frac{-ig_{\mu\nu}}{(p_1-p_3)^2}$$

The $g_{\mu\nu}$ contracts $\mu$ with $\nu$, turning $\gamma^\nu$ into $\gamma_\mu$:
**Multiply everything:**

$$-i\mathcal{M}_1 = (ig_e)[\bar{u}(3)\gamma^\mu u(1)] \cdot \frac{-ig_{\mu\nu}}{(p_1-p_3)^2} \cdot (ig_e)[\bar{u}(4)\gamma^\nu u(2)]$$

$$= \frac{-ig_e^2}{(p_1-p_3)^2}\,[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)] \tag{1}$$

Dividing both sides by $-i$:

**Amplitude $\mathcal{M}_1$**

$$\mathcal{M}_1 = -\frac{g_e^2}{(p_1-p_3)^2}\,[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]$$

The denominator $(p_1-p_3)^2$ will later be called $t$ (a Mandelstam variable).

**What Is This Object?**

$\mathcal{M}_1$ is a product of:

- $g_e^2/t$ — a number (coupling strength divided by propagator).
- $[\bar{u}(3)\gamma^\mu u(1)]$ — a **number** for each value of $\mu$ (it's a $1\times 4$ row times a $4\times 4$ matrix times a $4\times 1$ column $= 1\times 1$). This is the "current" at the top vertex.
- $[\bar{u}(4)\gamma_\mu u(2)]$ — another number for each $\mu$. This is the current at the bottom vertex.
- The $\mu$ index is summed over (Einstein convention), contracting the two currents.

So $\mathcal{M}_1$ is just a single complex number (for each choice of the four particle spins).

## §4.2 Diagram 2: By Analogy

$\mathcal{M}_2$ is the same as $\mathcal{M}_1$ with $p_3 \leftrightarrow p_4$ (and $s_3 \leftrightarrow s_4$):

**Amplitude $\mathcal{M}_2$**

$$\mathcal{M}_2 = -\frac{g_e^2}{(p_1-p_4)^2}\,[\bar{u}(4)\gamma^\mu u(1)][\bar{u}(3)\gamma_\mu u(2)]$$

The denominator $(p_1-p_4)^2$ will later be called $u$.

# Part II
# The Tools You Need

## §5 Mandelstam Variables

**What Are They?**

For any $2 \to 2$ process ($p_1 + p_2 \to p_3 + p_4$), define three numbers:

$$s \equiv (p_1 + p_2)^2 \qquad \text{(total energy squared in CM frame)}$$
$$t \equiv (p_1 - p_3)^2 \qquad \text{(momentum transfer, diagram 1)}$$
$$u \equiv (p_1 - p_4)^2 \qquad \text{(momentum transfer, diagram 2)}$$

These are **Lorentz invariants** — the same value in every reference frame.
**Why use them?** Because every dot product $p_i \cdot p_j$ can be written in terms of $s, t, u$. This means your final answer is automatically frame-independent.

### §5.1 Converting Dot Products (Massless Case)

Expand $(p_1 + p_2)^2$ using $p_i^2 = m^2 = 0$:

$$s = p_1^2 + 2p_1 \cdot p_2 + p_2^2 = 0 + 2p_1 \cdot p_2 + 0 \quad \Longrightarrow \quad p_1 \cdot p_2 = \frac{s}{2}$$

Similarly for the others. Here is the complete dictionary:

**Dot Product ↔ Mandelstam Dictionary (All Masses = 0)**

$$\begin{array}{lcl|lcl} p_1 \cdot p_2 & = & \dfrac{s}{2} & p_3 \cdot p_4 & = & \dfrac{s}{2} \\ p_1 \cdot p_3 & = & -\dfrac{t}{2} & p_2 \cdot p_4 & = & -\dfrac{t}{2} \\ p_1 \cdot p_4 & = & -\dfrac{u}{2} & p_2 \cdot p_3 & = & -\dfrac{u}{2} \end{array}$$

**Constraint:** $s + t + u = 0$, so $u = -s - t$.
**Propagators:** $(p_1 - p_3)^2 = t$ and $(p_1 - p_4)^2 = u$.

**Where Do the Minus Signs Come From?**

$t = (p_1 - p_3)^2 = -2p_1 \cdot p_3$ (since $p_1^2 = p_3^2 = 0$). In the CM frame, $t$ is always negative for scattering (the particles deflect, so the spatial part dominates). That's why $p_1 \cdot p_3 = -t/2$ comes out positive.

## §6 Casimir's Trick: Spin Sums → Traces (A Refresher)

**The Big Idea**

We want to compute $\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2$.
This sum runs over all spin states of all four particles (16 terms). Computing each one separately would be awful. Casimir's trick lets us do the *entire sum at once* by converting it into a trace of gamma matrices.
The key identity: when you sum a spinor bilinear over spins, the completeness relation replaces the outer product $u\bar{u}$ with a matrix:

$$\sum_{s=1}^{2} u^{(s)}(p)\,\bar{u}^{(s)}(p) = \not{p} + m$$

In our massless case ($m = 0$), this is simply $\not{p}$.

**How It Works in Practice**

When you encounter a product like $[\bar{u}(a)\Gamma_1 u(b)][\bar{u}(b)\Gamma_2 u(a)]$ and sum over the spins of $a$ and $b$:

1. Each $u(p)\bar{u}(p)$ pair gets replaced by $\not{p}$ (completeness).
2. The whole thing closes into a **trace**: $\text{Tr}[\Gamma_1 \not{p}_b \Gamma_2 \not{p}_a]$.
3. No spinors survive — everything is matrices and traces.

## §7 Contraction Theorems: The Tools for Evaluating Traces

Once we have traces, we need to evaluate them. The key tools are:

**The Contraction Theorems You'll Use Here**

**Rule 8′ — Two slashes inside:**

$$\gamma_\mu \not{a} \not{b} \gamma^\mu = 4(a \cdot b)$$

*In words:* Contracting the outer gammas around two slashed four-vectors collapses everything into 4 times the dot product. All gamma matrices disappear, leaving just a number.

**Rule 9′ — Three slashes inside:**

$$\gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\,\not{c} \not{b} \not{a}$$

*In words:* Contracting around three slashed quantities reverses their order and multiplies by $-2$.

**Basic trace theorem:**

$$\text{Tr}(\not{a} \not{b}) = 4(a \cdot b)$$

*In words:* The trace of two slashed quantities is 4 times their dot product.

**Why These Are So Powerful**

An 8-gamma-matrix trace looks terrifying. But contraction theorems let you shrink it step by step:
$8 \xrightarrow{\text{Rule } 9'} 6 \xrightarrow{\text{Rule } 8'} \text{number} \times 2 \xrightarrow{\text{basic trace}} \text{number}$
Two contractions and one trace evaluation. That's it.

# Part III
# The Calculation

## §8 The "Easy" Terms: $\langle|\mathcal{M}_1|^2\rangle$ and $\langle|\mathcal{M}_2|^2\rangle$

**Why These Are Easy**

$\mathcal{M}_1$ has the *exact same structure* as electron–muon scattering: two separate fermion currents connected by one photon. Griffiths already computed the spin-averaged $|\mathcal{M}|^2$ for that process (Eq. 7.129). We just set $m = M = 0$.
The key result from Eq. 7.129 (with $m = M = 0$):

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2g_e^4}{(p_1 \cdot p_3)^2}\left[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3)\right]$$

### §8.1 Translating to Mandelstam

Plug in the dictionary from Section 5.1:

$$\begin{aligned}(p_1 \cdot p_3)^2 &= (-t/2)^2 = t^2/4 \\ (p_1 \cdot p_2)(p_3 \cdot p_4) &= (s/2)(s/2) = s^2/4 \\ (p_1 \cdot p_4)(p_2 \cdot p_3) &= (-u/2)(-u/2) = u^2/4 \end{aligned} \tag{2}$$

Substituting:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2g_e^4}{t^2/4}\left[\frac{s^2}{4} + \frac{u^2}{4}\right] = \frac{8g_e^4}{t^2} \cdot \frac{s^2+u^2}{4}$$

**Direct Term**

$$\boxed{\langle|\mathcal{M}_1|^2\rangle = 2g_e^4\,\frac{s^2+u^2}{t^2}}$$

For $\langle|\mathcal{M}_2|^2\rangle$: since $\mathcal{M}_2$ is $\mathcal{M}_1$ with $p_3 \leftrightarrow p_4$, which swaps $t \leftrightarrow u$:

**Exchange Term**

$$\boxed{\langle|\mathcal{M}_2|^2\rangle = 2g_e^4\,\frac{s^2+t^2}{u^2}}$$

# §9 The Interference Term: $\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle$ — Step by Step

This is the new calculation. We'll go very slowly.

## §9.1 Step 1: Write $\mathcal{M}_1 \mathcal{M}_2^*$ in Terms of Spinors

We need:

$$\mathcal{M}_1 = -\frac{g_e^2}{t} \, [\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)] \tag{3}$$

$$\mathcal{M}_2^* = \left( -\frac{g_e^2}{u} \, [\bar{u}(4)\gamma^\nu u(1)][\bar{u}(3)\gamma_\nu u(2)] \right)^* \tag{4}$$

## §9.2 Step 2: Complex-Conjugate $\mathcal{M}_2$

**How to Conjugate a Spinor Bilinear**

For any $[\bar{u}(a)\,\Gamma\,u(b)]$, the complex conjugate swaps the spinors:

$$[\bar{u}(a)\,\Gamma\,u(b)]^* = \bar{u}(b)\,\bar{\Gamma}\,u(a)$$

For $\Gamma = \gamma^\mu$: $\bar{\Gamma} = \gamma^\mu$ (this holds for all QED vertex factors).
**Example:** $[\bar{u}(4)\gamma^\nu u(1)]^* = \bar{u}(1)\gamma^\nu u(4)$.
The coupling constant $g_e^2/u$ is real, so it passes through the conjugation unchanged.

Applying this to both bilinears in $\mathcal{M}_2$:

$$\mathcal{M}_2^* = -\frac{g_e^2}{u} \, [\bar{u}(1)\gamma^\nu u(4)][\bar{u}(2)\gamma_\nu u(3)]$$

So:

$$\mathcal{M}_1 \mathcal{M}_2^* = \frac{g_e^4}{tu} \underbrace{[\bar{u}(3)\gamma^\mu u(1)]}_{\text{A}} \underbrace{[\bar{u}(1)\gamma^\nu u(4)]}_{\text{B}} \underbrace{[\bar{u}(4)\gamma_\mu u(2)]}_{\text{C}} \underbrace{[\bar{u}(2)\gamma_\nu u(3)]}_{\text{D}} \tag{5}$$

## §9.3 Step 3: Identify the Spinor Loop

**The Key Observation**

Look at the spinor labels in A, B, C, D:

$$\text{A: } \bar{u}(\mathbf{3}) \to u(\mathbf{1}), \quad \text{B: } \bar{u}(\mathbf{1}) \to u(\mathbf{4}), \quad \text{C: } \bar{u}(\mathbf{4}) \to u(\mathbf{2}), \quad \text{D: } \bar{u}(\mathbf{2}) \to u(\mathbf{3})$$

The output of A ($u(1)$) feeds into the input of B ($\bar{u}(1)$). The output of B ($u(4)$) feeds into the input of C ($\bar{u}(4)$). And so on, forming a closed loop:

$$3 \to 1 \to 4 \to 2 \to 3$$

This is fundamentally different from the direct terms, where the spin sums factored into two independent loops. Here, all four particles are linked in **one** chain.

## §9.4 Step 4: Convert the Spin Sum to a Trace

When we sum over all spins, each $u(p)\bar{u}(p)$ pair in the chain gets replaced by $\not{p}$ (completeness relation with $m = 0$). Following the loop $3 \to 1 \to 4 \to 2 \to 3$ and writing down what you encounter at each step:

$$\not{p}_3 \quad \gamma^\mu \quad \not{p}_1 \quad \gamma^\nu \quad \not{p}_4 \quad \gamma_\mu \quad \not{p}_2 \quad \gamma_\nu$$

**Reading the Trace from the Loop**

At particle 3: write $\not{p}_3$ (from $u(3)\bar{u}(3) \to \not{p}_3$).
Then the gamma matrix at vertex A: write $\gamma^\mu$.
At particle 1: write $\not{p}_1$.
Then the gamma matrix at vertex B: write $\gamma^\nu$.
At particle 4: write $\not{p}_4$.
Then the gamma matrix at vertex C: write $\gamma_\mu$.
At particle 2: write $\not{p}_2$.
Then the gamma matrix at vertex D: write $\gamma_\nu$.
Close the loop $\to$ wrap everything in a trace:

$$\sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_2^* = \frac{g_e^4}{tu} \; \text{Tr}\Big[\not{p}_3 \, \gamma^\mu \, \not{p}_1 \, \gamma^\nu \, \not{p}_4 \, \gamma_\mu \, \not{p}_2 \, \gamma_\nu\Big]$$

With the spin-averaging factor of $1/4$:

$$\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{g_e^4}{4\,tu} \; \text{Tr}\Big[\not{p}_3 \, \gamma^\mu \, \not{p}_1 \, \gamma^\nu \, \not{p}_4 \, \gamma_\mu \, \not{p}_2 \, \gamma_\nu\Big]$$

This trace has **8 gamma matrices** inside it. Now we evaluate it.

## §9.5 Step 5: First Contraction — Kill the $\mu$ Index

We need to find $\gamma^\mu$ and $\gamma_\mu$ inside the trace and contract them. They're not adjacent — there are three objects between them: $\not{p}_1$, $\gamma^\nu$, $\not{p}_4$. The trace is cyclic, so we can rearrange:

$$\text{Tr}[\not{p}_3 \, \gamma^\mu \, \not{p}_1 \, \gamma^\nu \, \not{p}_4 \, \gamma_\mu \, \not{p}_2 \, \gamma_\nu] = \text{Tr}[\not{p}_2 \, \gamma_\nu \, \not{p}_3 \, \gamma^\mu \, \not{p}_1 \, \gamma^\nu \, \not{p}_4 \, \gamma_\mu]$$

(We cycled $\not{p}_2 \, \gamma_\nu$ from the end to the front — traces are cyclic so this doesn't change the value.)

Now $\gamma^\mu$ and $\gamma_\mu$ are at positions 4 and 8. The three objects between them (reading from $\gamma^\mu$ to $\gamma_\mu$) are: $\not{p}_1$, $\gamma^\nu$, $\not{p}_4$.

Apply **Rule** $9'$: $\gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2 \not{c} \not{b} \not{a}$:

$$\gamma^\mu \, \not{p}_1 \, \gamma^\nu \, \not{p}_4 \, \gamma_\mu = -2 \, \not{p}_4 \, \gamma^\nu \, \not{p}_1$$

**What Just Happened**

The three objects $\not{p}_1$, $\gamma^\nu$, $\not{p}_4$ got **reversed** to $\not{p}_4$, $\gamma^\nu$, $\not{p}_1$, and the whole thing picked up a factor of $-2$. The $\gamma^\mu$ and $\gamma_\mu$ are **gone** — consumed by the contraction.
We went from 8 gamma matrices inside the trace to 6.

The trace is now:

$$\mathrm{Tr}[\not{p}_2\,\gamma_\nu\,\not{p}_3\,(-2\,\not{p}_4\,\gamma^\nu\,\not{p}_1)] = -2\;\mathrm{Tr}[\not{p}_2\,\gamma_\nu\,\not{p}_3\,\not{p}_4\,\gamma^\nu\,\not{p}_1]$$

Cycle again to put $\gamma^\nu$ and $\gamma_\nu$ in convenient positions:

$$= -2\;\mathrm{Tr}[\not{p}_3\,\not{p}_4\,\gamma^\nu\,\not{p}_1\,\not{p}_2\,\gamma_\nu]$$

## §9.6 Step 6: Second Contraction — Kill the $\nu$ Index

Now $\gamma^\nu$ and $\gamma_\nu$ have two objects between them: $\not{p}_1$, $\not{p}_2$.

Apply **Rule** $8'$: $\gamma_\nu \not{a} \not{b} \gamma^\nu = 4(a \cdot b)$:

$$\gamma^\nu\,\not{p}_1\,\not{p}_2\,\gamma_\nu = 4(p_1 \cdot p_2) = 4 \cdot \frac{s}{2} = 2s$$

**What Just Happened**

The two slashed quantities $\not{p}_1$, $\not{p}_2$ collapsed into the number $4(p_1 \cdot p_2)$. The $\gamma^\nu$ and $\gamma_\nu$ are gone. The trace now contains only 2 gamma matrices:

$$-2 \cdot 2s \cdot \mathrm{Tr}[\not{p}_3\,\not{p}_4] = -4s\;\mathrm{Tr}[\not{p}_3\,\not{p}_4]$$

## §9.7 Step 7: Final Trace

The basic trace theorem gives:

$$\mathrm{Tr}[\not{p}_3\,\not{p}_4] = 4(p_3 \cdot p_4) = 4 \cdot \frac{s}{2} = 2s$$

So the full trace evaluates to:

$$-4s \cdot 2s = -8s^2$$

## §9.8 Step 8: Assemble the Result

$$\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{g_e^4}{4\,tu} \cdot (-8s^2) = \frac{-2g_e^4\,s^2}{tu}$$

Since this is a real number, the other cross term is the same: $\langle \mathcal{M}_1^* \mathcal{M}_2 \rangle = \langle \mathcal{M}_1 \mathcal{M}_2^* \rangle$.

**Interference Term — Final Answer**

$$\boxed{\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \langle \mathcal{M}_1^* \mathcal{M}_2 \rangle = \frac{-2g_e^4\,s^2}{tu}}$$

# §10 Combining All Four Terms

$$\langle |\mathcal{M}|^2 \rangle = \underbrace{\langle |\mathcal{M}_1|^2 \rangle}_{2g_e^4 \frac{s^2+u^2}{t^2}} + \underbrace{\langle |\mathcal{M}_2|^2 \rangle}_{2g_e^4 \frac{s^2+t^2}{u^2}} - \underbrace{\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle}_{\frac{-2g_e^4 s^2}{tu}} - \underbrace{\langle \mathcal{M}_1^* \mathcal{M}_2 \rangle}_{\frac{-2g_e^4 s^2}{tu}}$$

The interference terms come in with minus signs from $|\mathcal{M}_1 - \mathcal{M}_2|^2$, and the terms themselves are negative. Minus times minus gives plus:

$$-\left(\frac{-2g_e^4 s^2}{tu}\right) - \left(\frac{-2g_e^4 s^2}{tu}\right) = +\frac{4g_e^4 s^2}{tu}$$

**Full Spin-Averaged $|\mathcal{M}|^2$ for Møller Scattering ($m = 0$)**

$$\boxed{\langle|\mathcal{M}|^2\rangle = 2g_e^4\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu}\right]}$$

# §11 The Cross Section

## §11.1 What Is a Cross Section?

**Cross Section = "How Likely Is This Scattering?"**

The cross section $\sigma$ has units of area. It's the effective "target size" that one electron presents to the other. A bigger cross section means scattering is more likely.
$d\sigma/dt$ tells you how the scattering probability is distributed over different amounts of momentum transfer $t$.

## §11.2 The Formula

For massless $2 \to 2$ scattering:

$$\frac{d\sigma}{dt} = \frac{1}{16\pi s^2}\,\langle|\mathcal{M}|^2\rangle$$

But we have **identical particles** in the final state (two electrons), so we divide by $S = 2$ to avoid double-counting:

$$\frac{d\sigma}{dt} = \frac{1}{2}\cdot\frac{\langle|\mathcal{M}|^2\rangle}{16\pi s^2}$$

## §11.3 Plugging In

$$\frac{d\sigma}{dt} = \frac{1}{2}\cdot\frac{1}{16\pi s^2}\cdot 2g_e^4\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu}\right]$$

Use $g_e^2 = 4\pi\alpha$, so $g_e^4 = 16\pi^2\alpha^2$:

$$= \frac{1}{2}\cdot\frac{2\cdot 16\pi^2\alpha^2}{16\pi s^2}[\ldots] = \frac{\pi\alpha^2}{s^2}[\ldots]$$

**Final Answer: Differential Cross Section**

$$\boxed{\frac{d\sigma}{dt} = \frac{\pi\alpha^2}{s^2}\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu}\right]}$$

where $u = -s - t$ and $\alpha \approx 1/137$.

## §12 What You Just Learned: The General Pipeline

**The 5 Steps of Any QED Cross Section**

You just completed your first full QED cross section. Here's what you did, and you'll do the **exact same things** for every QED process (Bhabha, Compton, pair production, etc.):

**Step 1: Diagrams → Amplitude.** Use the Feynman rules to read off $\mathcal{M}$ for each diagram. Track which fermion line goes where, follow lines backwards, write down vertex factors and propagators.

**Step 2: Square.** If there are $N$ diagrams, you get $N^2$ terms. Direct terms ($|\mathcal{M}_i|^2$) and interference terms ($\mathcal{M}_i\mathcal{M}_j^*$).

**Step 3: Spin average.** Casimir's trick converts spin sums into traces. Direct terms give products of traces (one per fermion line). Interference terms give one big trace (a closed spinor loop).

**Step 4: Evaluate traces.** Contraction theorems (Rules 6–9) shrink the traces. Trace theorems (Rules 10–13) finish them off. Translate all dot products into Mandelstam variables.

**Step 5: Cross section formula.** Plug $\langle|\mathcal{M}|^2\rangle$ into $d\sigma/dt = \langle|\mathcal{M}|^2\rangle/(16\pi s^2)$. Include the identical-particle factor $S = 1/2$ if needed. Substitute $g_e^4 = 16\pi^2\alpha^2$. Done.

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Problem H4, Griffiths 7.37.
Sources: Griffiths Ch. 7, Thomson Ch. 6, Lancaster & Blundell Ch. 40.